

CREATE THREAD USING THREAD CLASS AND RUNNABLE INTERFACE

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1. Thread Class (java.lang.Thread)

- It is a built-in class that allows creating and managing threads.
- Used when we **extend** the Thread class.

Important Methods in Thread Class:

1. **start()** → Starts the thread and calls run() internally.
2. **run()** → Defines the task to be executed by the thread.
3. **sleep(ms)** → Makes the thread sleep for a given time.
4. **join()** → Waits for a thread to finish execution.
5. **yield()** → Gives CPU time to another thread.
6. **interrupt()** → Interrupts a sleeping or waiting thread.
7. **isAlive()** → Checks if the thread is still running.
8. **setPriority(int p)** → Sets the priority of a thread.
9. **getName()** / **setName(String name)** → Gets/Sets thread name.
10. **getState()** → Gets the current state of the thread.

2. Runnable Interface (java.lang.Runnable)

- It is an interface that is used when we implement Runnable.
- Does **not** have many methods like Thread, only one method:

Method in Runnable Interface:

1. **run()** → Contains the code that the thread will execute.

➤ Thread Class :

```
package com.string;
```

```
//Step 1: Extend Thread class
class MyThread extends Thread {
    // Step 2: Override run() method
    public void run() {
        System.out.println("Thread is running...");
    }
}

//Step 3 & 4: Create object and start the thread
public class Main {
    public static void main(String[] args) {
        MyThread t1 = new MyThread(); // Creating thread object
        t1.start(); // Starting the thread
    }
}
```

If I will start again the same thread twice then it will throw an exception. Because it already is in the dead state so if we try to alive that thread again so it is not possible .

➤ It will provide a runtime exception.

RUNNABLE:

```
package com.string;
```

```
class MyThread implements Runnable {
    public void run() {
        System.out.println("Thread is running...");
    }
}
```

```
public class Main {  
    public static void main(String[] args) {  
        MyThread t1 = new MyThread();  
        Thread thread = new Thread(t1);  
        thread.start();  
    }  
}
```

Question Arise : which is the better way ?

Ans: By implementing the runnable interaface.

Reason : When we working on big projects so that time many classes extends some other classes for reusability , so as we know that if we extend any class then multiple inheritance is not supported with classes.